Evaluating AI Agents

Adding Trajectory Evaluations





DeepLearning.AI

Talking Head

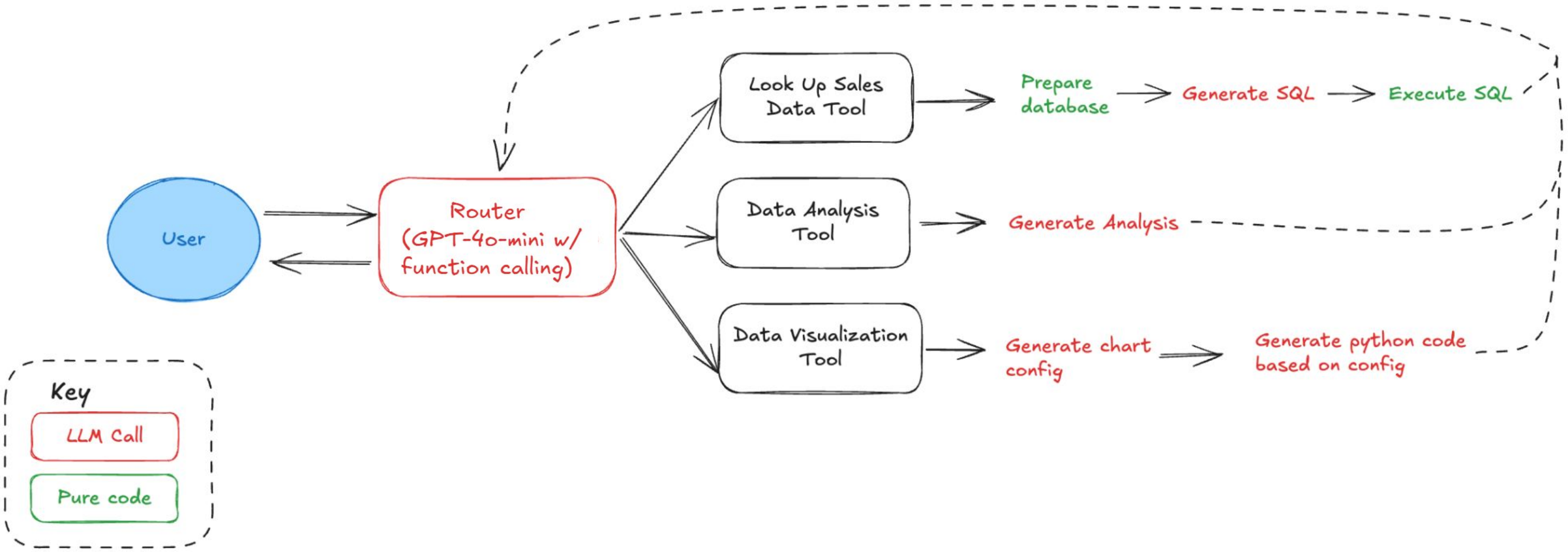
In this lesson:

- Agent Trajectories
- Calculating and Evaluating Convergence

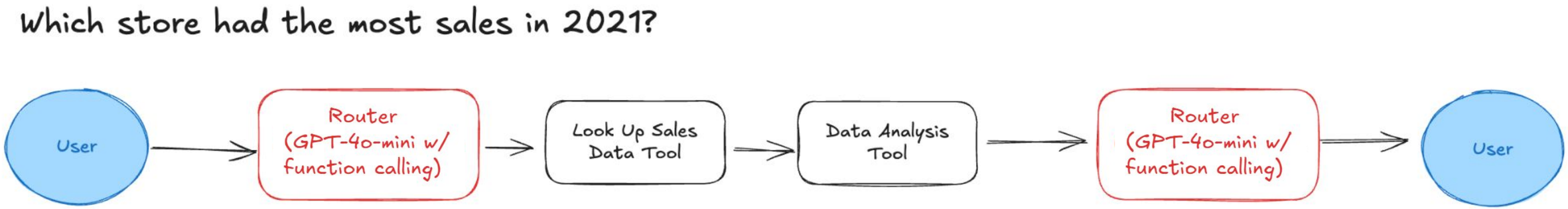
What is an Agent Trajectory?

- Trajectory: the path through router steps, tool calls, and logic steps that the agent took for a given input.

What is an Agent Trajectory?

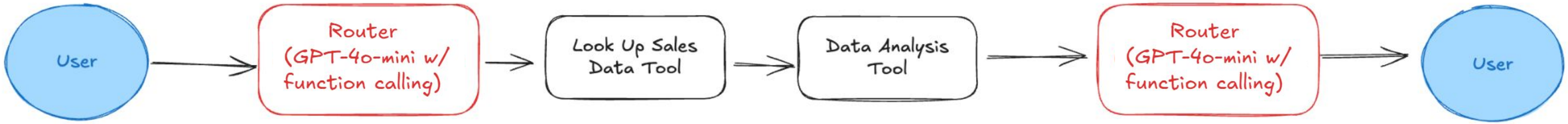


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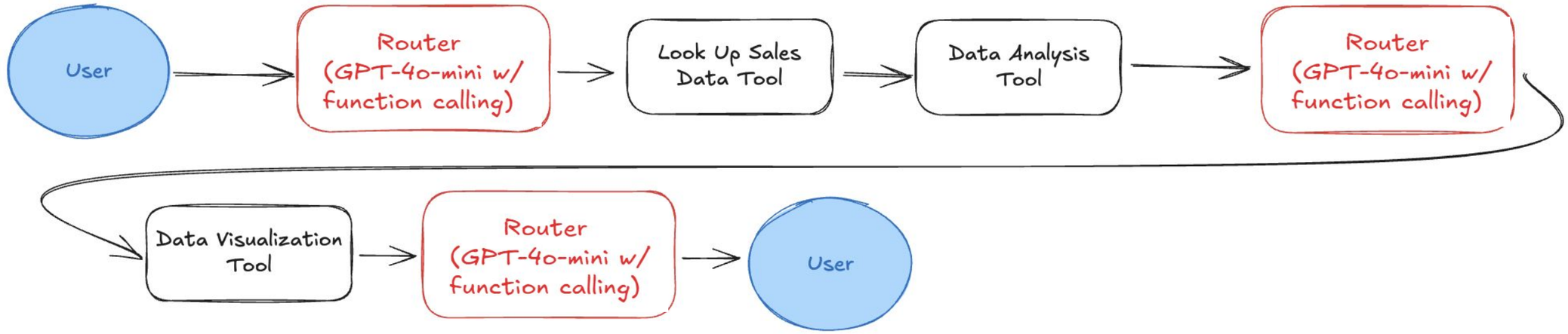


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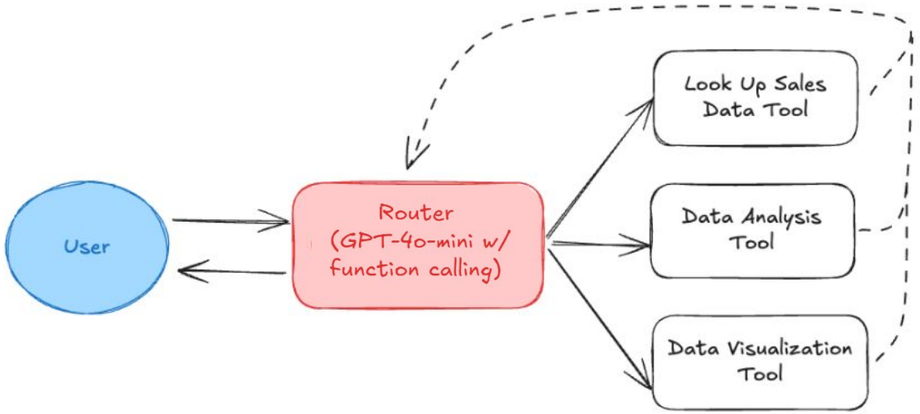
Which store had the most sales in 2021?

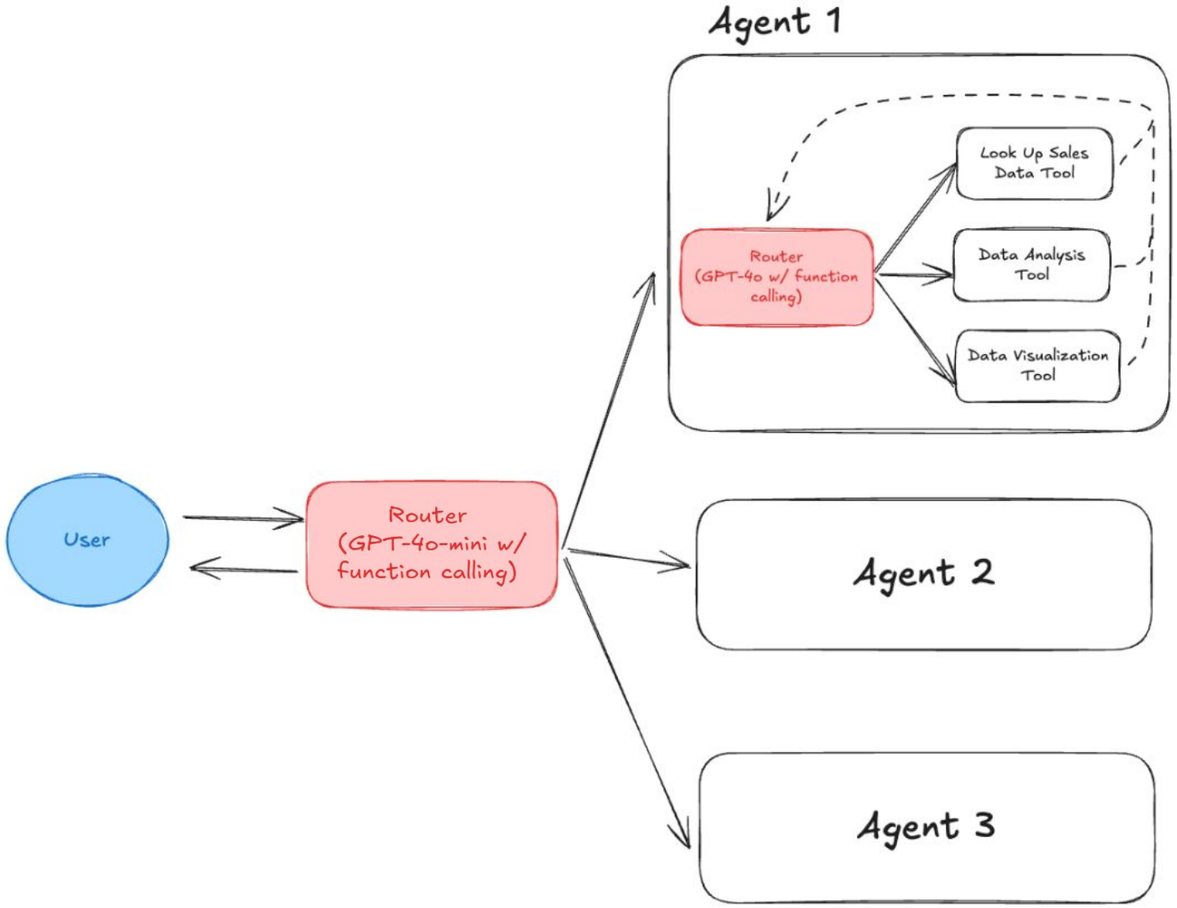


Plot daily sales volume over time



Trajectories get complicated quickly

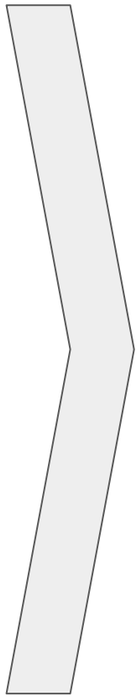




If your Agent's output is correct, does the Trajectory matter?

Yes, because efficiency matters!

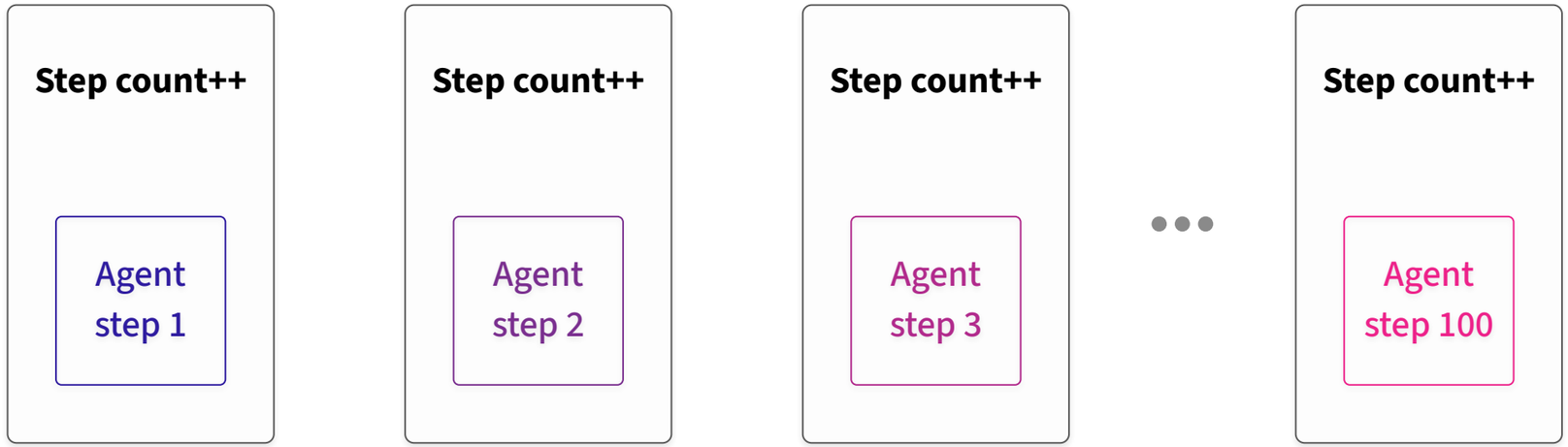
1. User input
2. Router
3. Database Lookup Tool
4. Router
5. Data Analyzer
6. Router
7. Database Lookup Tool
8. Router
9. Data Analyzer
10. Router
11. Output



1. User input
2. Router
3. Database Lookup Tool
4. Data Analyzer
5. Router
6. Output

Convergence

Convergence measures how closely your agent follows the optimal path for a given query.



How do you test for Convergence?

1. Run your agent on a set of similar queries
   - N: Total number of runs.
2. Record the number of steps taken for each
   - $S_{\text{agent},i}$: Steps taken by the agent in the i-th run.
3. Find the length of the optimal path
   - $S_{\text{optimal}} = \min(S_{\text{agent},1}, S_{\text{agent},2}, \ldots, S_{\text{agent},N})$ (minimum steps across all runs).
4. Calculate the **convergence score**:

$$\text{Overall Convergence Score} = \frac{\sum_{i=1}^{N} \min\left(1, \frac{S_{\text{optimal}}}{S_{\text{agent},i}}\right)}{N}$$

How do you test for Convergence?

Another way to think about a convergence score:

- What percentage of the time is your agent taking the optimal path, for a given set of inputs?
- Convergence score of 1 means that agent is taking the optimal path 100% of the time

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Note 1: convergence evaluations won't catch situations where an unnecessary step is taken by the agent every time.

Note 2: convergence evaluators must check that the agent fully completed a run

What you've learned:

1. What trajectory and convergence are, and why they're important to track
2. How to measure and calculate convergence